



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0055



Agricultural Machinery – Agricultural Trailer ACT 3 TONNE TRAILER

Test Report Number	2023 – 0055
Date of Issuance	March 7, 2023
Valid Until	March 7, 2028
Agricultural Machinery	Agricultural Trailer
Type	Semi-trailer, Hitched to four-wheel tractor
Brand	ACT
Model	3 TONNE TRAILER
Test Requested by	ACT (MACHINERIES & METAL CRAFT) CORPORATION Brgy. Turayong, Cauayan City, Isabela

SUMMARY

Construction Criteria	Requirement as per PAES 136:2004	AMTEC Test
Capacity, kg		
Gross Load	3560 ^a	4310 ^b
Payload	3000 ^a	3750
Overall width of the trailer, maximum, m	2.5	2.19
Overall length of the platform, maximum, m	6.0	2.98
Distance from the foremost end of the platform to the center of the hitch point of the pullbar, m	0.9-1.4	1.39
Overall height of the trailer, maximum, m	2.2	1.46
Overall height of the platform ^c , maximum, m	1.2	0.645
Distance from the rearmost end of the platform to the center of the wheel, m	0.992 ^d to 1.2	1.47
Wheel tread, mm	1500 ^a	1750
Wheel base, mm	NA	NA
Ground clearance, mm	410 ^a	317
Noise Level, dB(A), maximum	95 ^e	93.2

^a As specified by the requesting party

^b Equals to the measured payload plus unladen mass

^c Measured from the ground to the top of the flooring

^d One-third of the platform's length (semi-trailer)

^e Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Center, Department of Labor and Employment, 2013

Note: NA – Not Applicable

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	ACT (MACHINERIES & METAL CRAFT) CORPORATION
4. Tractor used	70.8 kW MASSEY FERGUSON 5709 Four-wheel Tractor
5. Number of operators	One
6. Handling and stability when turning	The trailer was stable when turning during the test operation.
7. Ability of the parking brake to hold the fully-laden trailer on 15° slope uphill or downhill	Not applicable
8. Applying force to hand brake lever	Not applicable
9. Visible deformation during loading capacity test	No visible deformation was observed during loading capacity test.
10. Safety features	The trailer is provided with taillights, an SMV emblem, and yellow reflectors.
11. Failure or abnormalities that may be observed on the trailer or its component parts during and after the operation	No failure or abnormalities were observed on the trailer or its component parts during and after the test operation.
12. The loading platform may be bare or provided with hinged or fixed sideboards.	The loading platform is equipped with hinged sideboards.
13. The trailer shall be provided with lashing hooks for tying down the load.	Provided
14. The trailer shall be provided with safety chain to match the trailer's gross load.	Not provided
15. When the trailer is fully loaded and hitched to the tractor, the platform shall be parallel to the ground.	The platform was parallel to the ground when the trailer is fully loaded and hitched to the tractor.

OBSERVATIONS (Cont'n)

16. The hitch point of the trailer shall be designed in accordance with the tractor drawbar requirements of four-wheel tractor given in PAES 118.	The hitch point of the trailer is compatible with that of the four-wheel tractor used. However, the drawbar requirement of the four-wheel tractor based on PAES 118 was not verified.
17. The trailer shall be fitted with suitable pneumatic wheels of adequate load carrying capacity.	The trailer is fitted with suitable pneumatic tires which were able to carry the load of the trailer during the load capacity test.
18. The trailer should be provided with spring loaded axles, if desired by the purchaser.	The trailer is not equipped with spring loaded axles.
19. The trailer may be provided with oscillating pullbar.	The trailer is provided with a fixed pullbar.
20. Trailer attached to a four-wheel tractor designed for more than 6 metric tons gross load shall be fitted with over-run brake. The over-run brake along with the tractor service brake shall be able to decelerate and stop the fully loaded tractor-trailer combination within a distance of 12 m when traveling at a maximum speed of 20 kph.	Not applicable
21. A parking brake shall be provided for all types of trailers. The parking brake shall hold a fully laden trailer on a 15° slope uphill or downhill. The maximum force to operate a hand brake lever shall not exceed 400 N. The parking brake should operate on the same drum and shoe or disc and pad as the service brake.	Parking brake was not provided.
22. The trailer shall be fitted with six amber (yellow) reflectors of not less than 75 mm diameter placed at a distance of not more than 150 mm inwards from extreme ends of rear and sides.	Six reflective stickers which are arrow-shaped are placed on the rear part of the trailer (Figure 1).
23. The trailer shall be fitted with tail lights and, when possible, with stop and turn signal lights.	The trailer is provided with tail, stop, and turn signal lights.
24. The semi-trailer for four-wheel tractors shall be provided with a park stand to keep it level when detached from the towing tractor.	A park stand is provided when the trailer is detached and parked.
25. A suitable unobstructed location shall be made available on the trailer chassis for fixing a lift jack when carrying out necessary repairs.	Lift jack points are provided under the chassis.
26. Agricultural trailer tires shall conform to ISO 4251 Parts 1 to 3.	The tires of the trailers were not verified if they conform to ISO 4251.

OBSERVATIONS (Cont'n)

27. All fasteners used in the trailer shall conform to PAES 311, PAES 312, PAES 313, PAES 314 and PAES 315.	The conformities to PAES of the fasteners used were not verified.
28. All parts requiring lubrication shall be provided with suitable fittings.	Not verified
29. The trailer shall be fitted with slow-moving vehicle (SMV) emblem. The emblem shall be located at the rear of the trailer with dimensional requirement as shown in Figure 4.	A slow-moving emblem (SMV) is provided at the rear of the tractor (Figure 2).
30. The trailer shall be free from manufacturing defects that may be detrimental to its operation.	No manufacturing defects were observed on the trailer during the test operation.
31. The trailer shall be free from sharp edges and surfaces that may injure the operator.	No sharp edges and surfaces were observed on the trailer during the test operation.
32. All wood or metal surfaces shall be covered with a coat of suitable preservative and an appropriate primer before painting.	All metal surfaces are painted.
33. Other remarks	Nameplate and other markings or labels are provided.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 8, 2023
Location of Test	Brgy. Tagaran, Cauayan City, Isabela 16°57'55.6" N, 121°45'49.1"E (Paved road) 16°58'05.7" N, 121°45'20.5"E (Bare dirt road)
Items Tested	Operator's Manual, Construction, Payload Test, Traveling Speed, Noise Level, and Fuel Consumption Rate

METHODS OF TEST

The agricultural trailer was tested based on PAES 137:2004 Agricultural Machinery – Agricultural Trailer – Methods of Test.

DESCRIPTION OF THE MACHINE

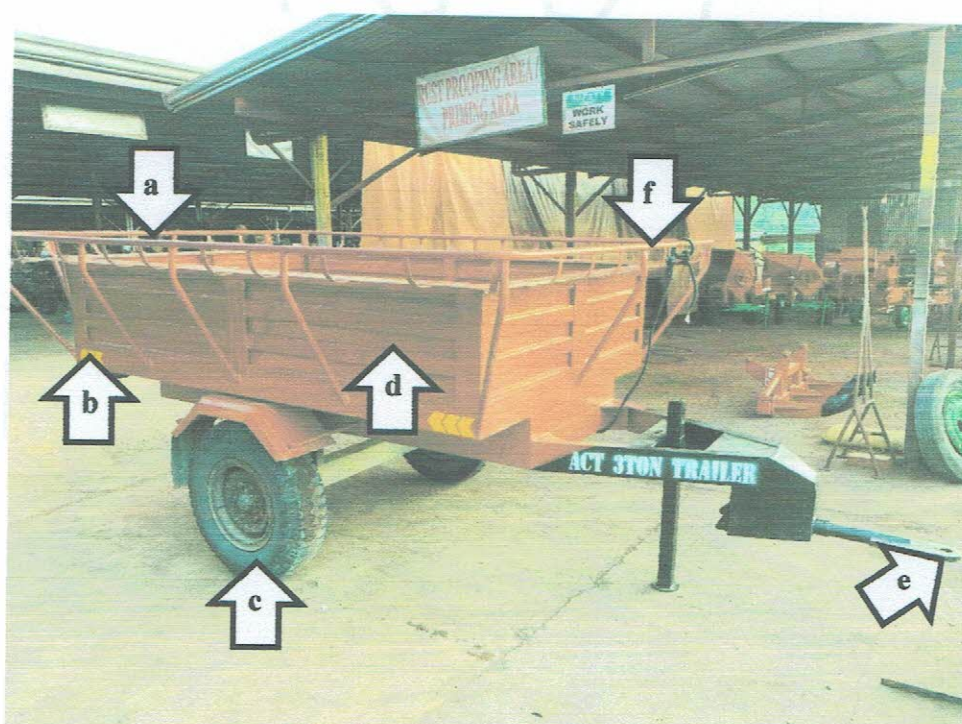


Figure 1. Parts of the ACT 3 TONNE TRAILER Agricultural Trailer hitched to 70.8 kW MASSEY FERGUSON 5709 Four-wheel Tractor includes the following: (a) Loading platform, (b) Yellow reflector, (c) Transport wheels, (d) Sideboard, (e) Hitch point, and (f) Extension for loading.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Overall dimensions and weight		
1.1 Trailer		
1.1.1 Length, mm	4120	4190
1.1.2 Width, mm	2190	2185
1.1.3 Height, mm	1470	1590
1.1.4 Unladen mass, kg	560	560
1.1.5 Payload, kg	3000	3750
1.1.6 Gross load, kg	3560	4310
1.2 Platform		
1.2.1 Material	G.I. sheet (Gauge 16)	Flat bar (t=4.00 mm), Angle bar (t=4.54 mm), B.I. Pipe (t=26.54 mm), B.I. Sheet (t=1.15 mm)
1.2.2 Dimension, L × W × H, mm	2340 × 1539 × 496	2975 × 2185 × 645
1.2.3 Ground clearance, mm	319	317
1.2.4 Overhang (front), mm	1470	313
1.2.5 Overhang (rear), mm	1200	310
1.2.6 Overhang (side), mm	ND	332
1.3 Pull bar		
1.3.1 Material	CR Shafting (50.8 mm diameter) MS Plate (t = 20 mm)	C-bar and Square bar
1.3.2 Length, mm	1460	1385
1.3.3 Thickness of clevis, mm	19.51	C-bar (t=5.80 mm) Tow eye (t=20.12 mm)

^a The data were obtained from the operator's manual and specifications sheet submitted by the requesting party.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
1.4	Tires		
1.4.1	Tire specification		
1.4.1.1	Size, W × D, mm	177.80 × 381 (7.00 – 15 LT)	177.80 × 381
1.4.1.2	Ply rating	12	Bias-ply
1.4.1.3	Inflation pressure, kg/m ²	100	80
1.4.2	Wheel base, mm	NA	NA
1.4.3	Wheel tread, mm	1490	1750
1.5	Tractor-trailer combination		
1.5.1	Overall length, mm	ND	7757
1.6	Maximum dumping angle	ND	NM
1.7	Safety/precautionary markings	with SMV emblem and yellow reflectors	with SMV emblem and yellow reflectors

^a The data were obtained from the operator's manual and specifications sheet submitted by the requesting party.
 Note: NA – Not Applicable, ND – No Data, NM – Not Measured

RESULTS

Item	Data
1 Test conditions	
1.1 Location	Brgy. Tagaran, Cauayan City, Isabela 16°57'55.6" N, 121°45'49.1"E (Paved road) 16°58'05.7" N, 121°45'20.5"E (Bare dirt road)
1.2 Type of ground surface	Paved and bare dirt roads
2 Payload test	
2.1 Paved road	
2.1.1 Tractor gear setting	3 rd Gear - High
2.1.2 Load applied ^a , kg	3750
2.1.3 Traveling speed, kph	17.00
2.1.4 Braking distance, m	NA ^b
2.1.5 Noise level, dB(A)	93.2
2.1.6 Fuel consumption of the four-wheel tractor used, L/h	4.74
2.1.7 Operating time, h	3.00
2.2 Bare dirt road	
2.2.1 Tractor gear setting	3 rd Gear - High
2.2.2 Load applied ^a , kg	3750
2.2.3 Traveling speed, kph	9.40
2.2.4 Braking distance, m	NA ^b
2.2.5 Noise level, dB(A)	88.7
2.2.6 Fuel consumption of the four-wheel tractor used, L/h	1.89
2.2.7 Operating time, h	3.00
3 Parking brake test	NA ^b

^a Equal to 125% of the declared trailer's payload

^b The trailer is not equipped with an over-run, service, and parking brakes.

Note: NA – Not applicable

OBSERVATIONS

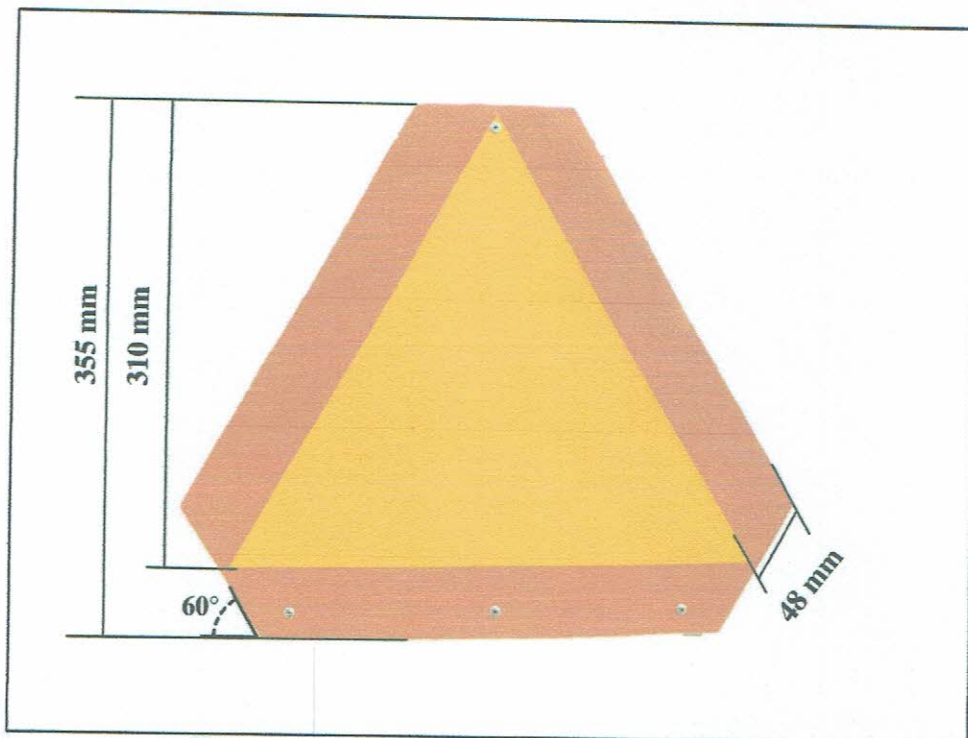


Figure 2. Dimensions of the SMV Emblem.

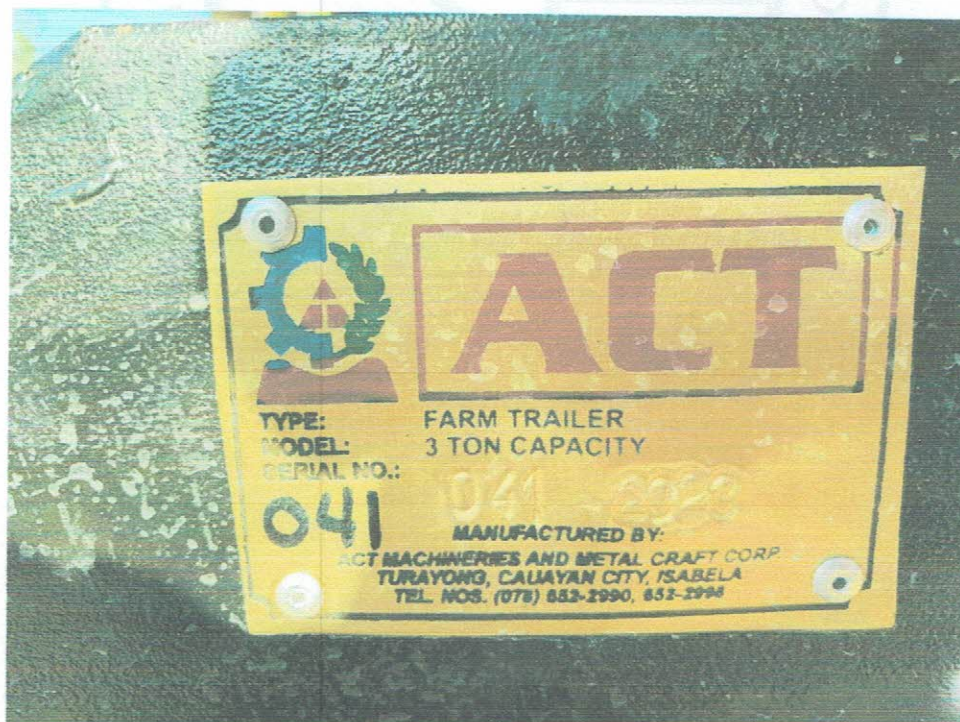


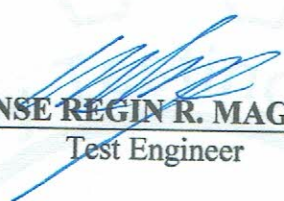
Figure 3. Nameplate of the machine.

Tested by:

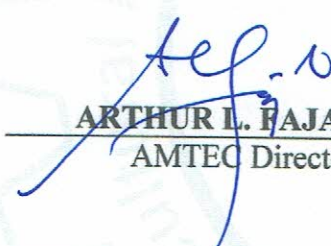

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